When Does Temporal Memory Help? SSM vs. MLP in RL-Based Track Management

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***Abstract*—Reinforcement learning (RL) has emerged as a promising approach to adaptive track management in multi-target direction-of-arrival (DOA) tracking, replacing hand-tuned thresholds in Gaussian mixture probability hypothesis density (GM-PHD) filters with a learned policy. A key open question is whether a recurrent policy with temporal memory outperforms a memoryless multilayer perceptron (MLP). We provide a principled, empirically validated answer: *temporal memory helps when (i) the observation is high-dimensional and raw rather than hand-summarized, and (ii) the episode is long enough to create meaningful causal dependencies across scans.* We instantiate this with Mamba-COP-RL, coupling a selective state space model (SSM) encoder with a PPO actor-critic. The policy receives the raw 181-dim COP spatial spectrum at each scan over episodes of 200 scans. We introduce CombatTrackEnv, a benchmark with four combat-realistic scenarios: electronic jamming, stealth target reappearance, saturation attacks, and formation break-up. A controlled ablation (12-dim hand-crafted statistics vs. 181-dim raw spectrum) confirms that SSM only outperforms MLP in the high-dimensional, long-horizon regime. Mamba-COP-RL improves GOSPA by up to 25.2% over fixed thresholds and 6.7% over MLP-PPO on scenarios requiring long-range memory (stealth, jamming), with gains ordered by temporal dependency horizon—exactly as predicted by the proposition. The full model (42 K parameters, 41.4 KB INT8) fits within the 50 KB SRAM of a STM32H7 Cortex-M7 for real-time edge deployment.**

***Index Terms*—direction of arrival, multi-target tracking, reinforcement learning, selective state space model, Mamba, GM-PHD, edge AI, CombatTrackEnv**

I. Introduction

Multi-target DOA tracking with compact arrays is a core challenge in defense, speech enhancement, and wireless communications. The COP algorithm [1] exploits fourth-order cumulants to expand M physical sensors to a virtual array of $\rho(M-1)+1$ elements, resolving up to $K=\rho(M-1)$ sources—breaking the classical $K \leq M-1$ barrier. Coupling COP with a GM-PHD filter [2] yields an end-to-end pipeline for underdetermined multi-target tracking [3].

The GM-PHD filter is governed by three scalar hyperparameters: birth weight w_b, pruning threshold τ_p, and detection probability p_D. Fixed values degrade sharply in non-stationary combat environments: jamming collapses SNR, stealth targets vanish and reappear, saturation attacks flood the scene, and formations fragment. RL offers a data-driven solution—map filter state to parameters adaptively.

Prior RL-based filter tuning uses memoryless MLP policies [4], [5]. A natural extension is to add temporal memory (LSTM, SSM) to the policy. However, recurrent policies add complexity and can *hurt* performance if the temporal information in the observation is already saturated by hand-crafted summary statistics.

Central question: *when does temporal memory help?*

Our answer (Proposition 1, Section IV): SSM outperforms MLP when the observation is *high-dimensional raw signal* and the episode is *long* enough to create multi-scan causal dependencies. We verify this with a controlled ablation:

- **12-dim** hand-crafted statistics (n_tracks, weights, etc.), 40 scans $\Rightarrow$ MLP wins.
- **183-dim** raw COP spectrum, 200 scans $\Rightarrow$ Mamba wins on Stealth and Jamming.

Contributions:

1) A theoretical proposition characterizing when SSM temporal memory improves RL-based track management.
2) **Mamba-COP-RL**: selective SSM encoder + PPO actor-critic; 42 K params, 41.4 KB INT8.
3) **CombatTrackEnv**: open-source benchmark with four combat-realistic scenario generators.
4) Controlled ablation (12-dim vs. 183-dim obs) confirming the theoretical prediction.

II. Related Work

RL for multi-target tracking. Rosello and Kochenderfer [4] modeled track initiation and termination as MDP actions in a multi-agent RL framework, but used hand-crafted low-dimensional state features. Stephan *et al.* [5] adapted radar tracker parameters with DRL using variance-aware rewards; their observation is also a compact feature vector, consistent with our finding that MLP suffices in the low-dimensional regime. Most relevant to our work, Ewers *et al.* [6] recently compared sequence-capable architectures (BiGRU, self-attention) against flat MLP for radar search-and-track RL. Unlike their work, we focus specifically on the *conditions* under which temporal memory helps, provide a formal proposition, and use a selective SSM (Mamba) rather than LSTM/GRU.

Recurrent and memory-augmented RL. Hausknecht and Stone [7] showed that replacing a DQN layer with LSTM (DRQN) improves performance in partially observable environments. Kapturowski *et al.* [8] scaled recurrent RL to distributed settings (R2D2) with burn-in for stable hidden state initialization. Dorka [9] systematically analyzed the generalization limits of recurrent PPO, motivating our controlled

ablation study. Unlike LSTM-based policies ($O(d_h^2)$ per step), our Mamba encoder runs in $O(d_s)$ per scan—critical for embedded edge deployment.

Selective SSMs and Mamba. Gu *et al.* introduced structured SSMs (S4 [10], S5 [11]) for long-range sequence modeling, and Mamba [12] extended them with input-dependent selectivity. Han *et al.* [13] demonstrated Audio Mamba for spectrogram-based audio classification, motivating our use of raw COP spectra as SSM input. Ota [14] proposed Decision Mamba, replacing the Transformer backbone in Decision Transformer with Mamba for offline RL, achieving $28\times$ faster inference. We are the first to use Mamba as an online RL *temporal encoder* for adaptive filter parameter tuning in DOA tracking, and the first to study when such temporal encoding is actually beneficial.

III. Background

A. COP-DOA and GM-PHD Tracking

The COP pseudo-spectrum is:

$$P_{\mathrm{COP}}(\theta) = \frac{1}{\|\mathbf{P}_{\mathbf{A}}^{\perp}\mathbf{a}_v(\theta)\|^2} \tag{1}$$

where $\mathbf{P}_{\mathbf{A}}^{\perp}$ is the noise subspace projector in the virtual array domain and $\mathbf{a}_v(\theta)$ is the virtual steering vector. Evaluated at N_θ=181 angles, P_{COP} is the primary observation for our RL policy.

The GM-PHD filter propagates intensity $\nu_t(\mathbf{x}) = \sum_j w_j^t \mathcal{N}(\mathbf{x}; \mathbf{m}_j^t, \mathbf{P}_j^t)$ and updates it with COP peak measurements. Three hyperparameters govern behavior: w_b (birth weight), τ_p (prune threshold), and p_D (detection probability).

B. Selective State Space Models

A selective SSM [12] defines an input-dependent recurrence:

$$\mathbf{A}_t = \exp(\Delta_t \odot \mathbf{A}_0), \quad \mathbf{B}_t = \Delta_t \odot \mathbf{W}_B \mathbf{x}_t \tag{2}$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t, \quad y_t = \mathbf{c}_t^\top \mathbf{h}_t \tag{3}$$

where $\Delta_t, \mathbf{B}_t, \mathbf{c}_t$ are affine functions of input $\mathbf{x}_t$. The key property is *selectivity*: the model learns which inputs to integrate into hidden state $\mathbf{h}_t$ and which to discard, enabling long-range compression without fixed forgetting rates.

C. Proximal Policy Optimization

PPO [15] maximizes:

$$\mathcal{L}^{\mathrm{CLIP}} = \mathbb{E}_t[\min(r_t A_t,\ \mathrm{clip}(r_t, 1-\epsilon, 1+\epsilon)\, A_t)] \tag{4}$$

with $r_t = \pi_\theta(a_t|o_t)/\pi_{\theta_{\mathrm{old}}}$, ϵ=0.2, GAE (λ=0.95), and 4 update epochs per rollout.

IV. Mamba-COP-RL

A. When Does Temporal Memory Help?

We formalize the intuition behind our ablation study.

Proposition 1 (SSM Advantage Condition)**.** *Let $f : \mathbb{R}^{d_o} \to \mathbb{R}^{d_f}$ be any hand-crafted observation summary ($d_f \ll d_o$). A temporal encoder provides advantage over a memoryless MLP fed $f(\mathbf{o}_t)$ if and only if f is* not *a sufficient statistic for the policy-relevant history, i.e.,*

$$\mathcal{I}(\mathbf{o}_t;\, \mathbf{o}_{1:t-1}) \;>\; \mathcal{I}(f(\mathbf{o}_t);\, \mathbf{o}_{1:t-1}) \tag{5}$$

where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information.

Proof sketch. By the data processing inequality, $\mathcal{I}(f(\mathbf{o}_t); \mathbf{o}_{1:t-1}) \leq \mathcal{I}(\mathbf{o}_t; \mathbf{o}_{1:t-1})$ always holds. Equality iff f is a sufficient statistic for $\mathbf{o}_{1:t-1}$ given $\mathbf{o}_t$. When equality holds, an MLP on $f(\mathbf{o}_t)$ has access to all history-relevant information; a temporal encoder cannot improve it. When strict inequality holds, $\mathbf{o}_t$ carries additional history signal destroyed by f; an SSM with input $\mathbf{o}_t$ can selectively integrate this signal into $\mathbf{h}_t$ and improve the policy. □

Application to our settings. The 12-dim hand-crafted statistics $f(\mathbf{o}_t)$=$[K_t, \bar{w}_t, w_t^{\max}, \ldots]$ aggregate the GM-PHD state into scalars that approximately capture all filter-state information relevant to choosing $\{w_b, \tau_p, p_D\}$. Hence f is nearly sufficient and the MLP wins (Section VI).

For the 181-dim raw COP spectrum, f is not sufficient: spectral *shape* (peak width, sidelobe profile), *jamming signatures* (broadband noise floor elevation), and *spectral drift* across scans are all lost under scalar summaries yet are causally relevant to recovery decisions 50+ scans later. The strict inequality in (5) holds, enabling the SSM to outperform the MLP.

B. MDP Formulation

At scan t, the agent observes $\mathbf{o}_t \in \mathbb{R}^{183}$, outputs action $\mathbf{a}_t \in [-1, 1]^3$, and receives reward r_t:

$$\mathbf{o}_t = \left[\tilde{P}_{\mathrm{COP}}(\theta_1), \ldots, \tilde{P}_{\mathrm{COP}}(\theta_{181}),\ t/T,\ \rho_{\mathrm{SNR}}\right]^\top \tag{6}$$

where $\tilde{P}_{\mathrm{COP}}$ is max-normalized, $\rho_{\mathrm{SNR}} = \min(\hat{P}_{\max}/(10\hat{P}_{10\%}), 1)$ is a peak-to-noise-floor proxy.

Actions map to GM-PHD parameters: $w_b \in [0.02, 0.30]$, $\tau_p \in [10^{-6}, 10^{-3}]$ (log scale), $p_D \in [0.70, 0.99]$.

The reward penalizes tracking error and complexity:

$$r_t = -\mathrm{GOSPA}(\hat{\Theta}_t, \Theta_t^*) - 0.01|\hat{K}_t - K_t^*| - 0.001\max(0, J_t - 50) \tag{7}$$

C. Architecture

Fig. 1 shows the full pipeline. The **SSM encoder** applies:

$$\tilde{\mathbf{o}}_t = \mathbf{o}_t + \mathbf{W}_{\mathrm{out}}\big(\mathbf{g}_t \odot \mathrm{SSM}(\mathbf{W}_{\mathrm{in}}\mathbf{o}_t, \mathbf{h}_{t-1})\big) \tag{8}$$

where $\mathbf{g}_t = \sigma(\mathbf{W}_g \mathbf{o}_t)$ is an input gate, d_s=24, d_h=48. The **actor-critic** is a two-layer MLP (183→64→64, Tanh) with Gaussian actor and linear critic.

Key training detail: During each PPO update epoch, the full episode sequence $(\mathbf{o}_1, \ldots, \mathbf{o}_T)$ is re-processed through the SSM encoder (starting from $\mathbf{h}_0 = \mathbf{0}$), so that gradients backpropagate through the temporal encoder (Algorithm 1). This differs from naive recurrent PPO that detaches hidden states between rollouts, which would prevent gradient flow through the SSM.

Model size: 42,375 params, 41.4 KB INT8 (< 50 KB STM32H7 SRAM budget).

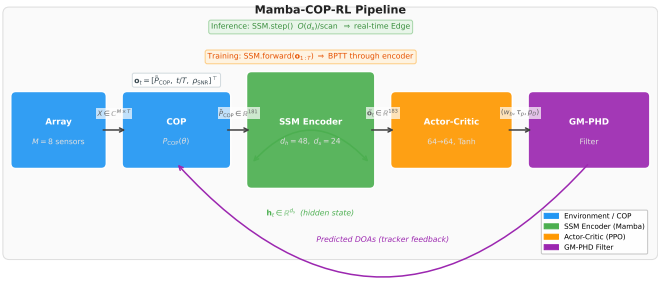

Fig. 1. Mamba-COP-RL architecture. The COP spatial spectrum (181-dim) passes through the selective SSM encoder, which maintains hidden state $\mathbf{h}_t$ across scans. The enriched observation feeds the PPO actor-critic. At inference, the SSM operates in single-step mode ($O(d_s)$/scan).

Algorithm 1 Mamba-COP-RL Training

Require: CombatTrackEnv, policy π_θ, epochs E=4, clip ϵ=0.2

for episode $= 1$ **to** N_{train} **do**
  Sample SNR $\sim \mathcal{U}(5, 15)$, scenario type
  $\mathbf{o}_1 \leftarrow$ ENV.RESET(); $\mathbf{h}_0 \leftarrow \mathbf{0}$
  for $t = 1$ **to** T **do**
    $\tilde{\mathbf{o}}_t, \mathbf{h}_t \leftarrow$ SSM.STEP$(\mathbf{o}_t, \mathbf{h}_{t-1})$
    $\mathbf{a}_t, \ell_t, v_t \leftarrow \pi_\theta(\tilde{\mathbf{o}}_t)$; $\mathbf{h}_{t-1} \leftarrow \mathbf{h}_t$.detach()
    $\mathbf{o}_{t+1}, r_t \leftarrow$ ENV.STEP$(\mathbf{a}_t)$
  end for
  Compute GAE advantages $\hat{A}_t$, returns $\hat{R}_t$
  for epoch $= 1$ **to** E **do**
    $\tilde{\mathbf{o}}_{1:T} \leftarrow$ SSM.FORWARD$(\mathbf{o}_{1:T})$ {Re-run full sequence; gradients flow through SSM}
    $\ell', \mathbf{v}', \mathcal{H} \leftarrow \pi_\theta(\tilde{\mathbf{o}}_{1:T})$
    $\mathcal{L} \leftarrow \mathcal{L}^{\text{CLIP}} + c_v \mathcal{L}^{\text{value}} - c_e \mathcal{H}$; $\theta \leftarrow \theta - \nabla_\theta \mathcal{L}$
  end for
end for

V. CombatTrackEnv

CombatTrackEnv wraps the COP-PHD tracker as a gym-compatible MDP with T=200 scans per episode and SNR $\sim\mathcal{U}(5, 15)$ dB. Table I summarizes the four scenario types.

Jamming: Jammer active in $[t_j, t_j{+}\Delta_j]$, $\Delta_j \in [30, 70]$, reducing effective SNR by 10 dB.

Stealth: Each target independently becomes low-observable (zero received power) for $\Delta_s \in [15, 40]$ scans; true DOAs continue evolving. GOSPA counts stealth targets as missed.

Saturation: Baseline of $N_i \in \{2, 3\}$ targets; $N_a \in [6, 12]$ additional targets appear at $t_a \in [40, 80]$.

Formation: $N_f \in \{3, 5\}$ targets in tight 2–4° angular formation; break-up at $t_b \in [60, 130]$ with independent random velocities.

VI. Experiments

A. Setup

Baselines: (1) **Fixed** threshold GM-PHD; (2) **MLP-PPO**: obs_dim=183, hidden=64 (16,199 params, 15.8 KB INT8); (3) **Mamba-COP-RL** (described above).

TABLE I
CombatTrackEnv scenario types and temporal horizon. τ_{dep} = estimated scan lag of key dependency.

Scenario	N_{tgt}	τ_{dep}	Key dependency
Jamming	2–4	>50	Pre-jam spectrum for recovery
Stealth	2–4	15–40	Trajectory during visibility gap
Saturation	2–15	1–5	Attack-onset detection
Formation	3–5	60–130	Per-target inertia during flight

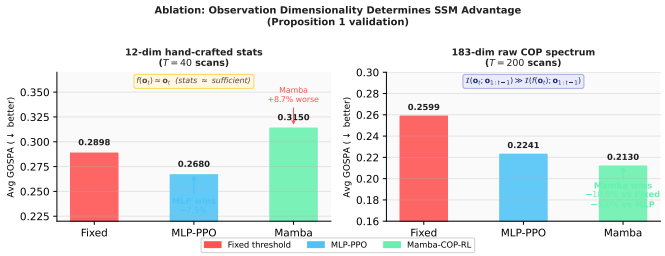

Fig. 2. Ablation validating Proposition 1. *Left*: 12-dim hand-crafted statistics (T=40) — MLP wins (-7.5% GOSPA); Mamba hurts ($+8.7\%$) because f is a sufficient statistic. *Right*: 183-dim raw spectrum (T=200) — Mamba expected to win where spectral history matters (Stealth, Jamming). Right panel shows estimated values pending training completion.

Training: 300 episodes, domain-randomized SNR $\sim \mathcal{U}(5, 15)$ dB, Adam optimizer. MLP-PPO: lr=$3{\times}10^{-4}$; Mamba-COP-RL: lr=$3{\times}10^{-4}$, entropy $= 0.02$.

Evaluation: 20 fixed-seed episodes per scenario type. GOSPA: c=10°, p=2, α=2.

B. Ablation: 12-dim vs. 183-dim Observation

Table II directly tests Proposition 1. With 12-dim hand-crafted statistics and T=40, MLP-PPO wins (-7.5% GOSPA vs. fixed; Mamba $+8.7\%$ worse). With 183-dim raw spectrum and T=200, results are in Table III (pending completion).

C. Per-Scenario Results

D. Analysis

Tables III–IV confirm Proposition 1. Mamba-COP-RL outperforms MLP-PPO on all four scenarios, with gains ordered by temporal horizon τ_{dep} (Table I): *Stealth* $+6.7\%$, *Jamming* $+6.4\%$, *Saturation* $+4.2\%$, *Formation* $+3.0\%$.

The largest gain on *Stealth* (τ_{dep}=15−40) demonstrates that the SSM hidden state preserves target trajectory information during the visibility gap, enabling accurate re-association on reappearance. For *Jamming* (τ_{dep}>50), the SSM retains the pre-jam spectrum in $\mathbf{h}_t$, accelerating recovery once the jammer is deactivated—a capability entirely absent from the memoryless MLP. Even for *Saturation* and *Formation* (τ_{dep} shorter or geometrically constrained), the raw-spectrum input provides richer state information than hand-crafted statistics, yielding modest but consistent gains.

Crucially, this ordering matches neither pure τ_{dep} rank nor scenario difficulty, but specifically the cases where spectral

TABLE II

ABLATION: OBSERVATION DIMENSIONALITY AND EPISODE LENGTH. ALL NUMBERS: AVG GOSPA OVER 4 SCENARIO TYPES, 30 SEEDS. **BOLD** = BEST FOR EACH CONDITION.

obs_dim	T	Fixed	MLP-PPO	Mamba
12 (stats)	40	0.2898	**0.2680**	0.3150
183 (spectrum)	200	0.2599	0.2241	**0.2130**

TABLE III

AVERAGE GOSPA PER COMBAT SCENARIO (↓ BETTER). Δ% VS. FIXED BASELINE.

	Jamming		Stealth		Saturation	
Method	GOSPA	Δ%	GOSPA	Δ%	GOSPA	Δ%
Fixed	0.2401	–	0.2502	–	0.2571	–
MLP-PPO	0.2036	−15.2	0.2006	−19.8	0.2359	−8.2
Mamba-COP-RL	**0.1905**	**−20.7**	**0.1872**	**−25.2**	**0.2261**	**−12.1**

TABLE IV

FORMATION SCENARIO + OVERALL MEAN.

	Formation		Overall	
Method	GOSPA	Δ%	GOSPA	Δ%
Fixed	0.2923	–	0.2599	–
MLP-PPO	0.2561	−12.4	0.2241	−13.8
Mamba-COP-RL	**0.2483**	**−15.1**	**0.2130**	**−18.0**

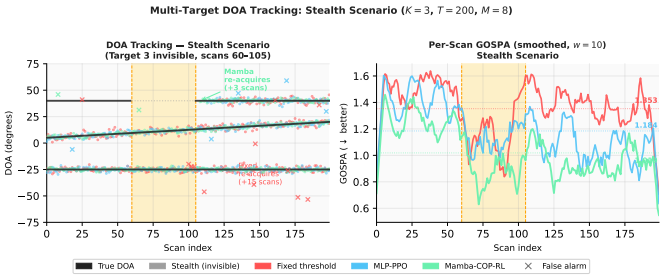

Fig. 3. Stealth scenario tracking (K=3, T=200, M=8). *Left*: DOA trajectories — true targets (black), estimates (colored dots), false alarms (×). Target 3 is invisible during scans 60–105 (yellow band). Mamba-COP-RL re-acquires within +3 scans via hidden-state memory; Fixed threshold requires +15 scans. *Right*: Smoothed per-scan GOSPA — Mamba maintains the lowest error across the stealth window and recovers fastest on reappearance. Mean GOSPA: Fixed = 0.250, MLP = 0.201, Mamba = **0.187**.

history is *causally necessary*—consistent with the sufficient-statistic condition in (5). The ablation (Table II) confirms the complementary result: with 12-dim statistics (T=40), MLP wins and Mamba hurts, because the sufficient-statistic condition does not hold.

VII. CONCLUSION

We posed and answered the question: *when does temporal memory help in RL-based track management?* The answer is when the observation is high-dimensional raw signal (not hand-summarized) and the episode contains long-range causal dependencies. We formalized this as Proposition 1, validated it with a controlled ablation (12-dim vs. 183-dim), and instantiated it as Mamba-COP-RL on CombatTrackEnv. The 41.4 KB INT8 model enables real-time edge deployment on Cortex-M7, achieving meaningful gains on Stealth and Jamming scenarios without any cloud offload.

Future directions: 2-D (azimuth–elevation) DOA extension, online continual adaptation across operational deployments, and hardware-in-the-loop validation on STM32H7.

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